American International University-Bangladesh (AIUB)

Faculty of Science & Technology (FST)

Department of Computer Science

Programming In Python

Mid-Term Project Report

On Maze Runner Game

Spring 2025-2026

Section: D

Group: 12

SL #	Student Name & ID	Contributions
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Title : Game Development using Pygame (Maze Runner)

Setup Instructions

1. Install Python (version 3.x recommended).
2. Install required library:

   pip install pygame
3. Ensure the structure is correct:
   a. main.py
   b. game.py
   c. settings.py
   d. storage.py
   e. entities/ (player.py, maze.py, coin.py)

Run Steps

1. Open terminal / command prompt.
2. Navigate to project folder:

   cd project_folder_name
3. Run the game:

   python main.py
4. The game window will open and start.

Keyboard Control

Use arrow keys:

Up Arrow- Move up
Down Arrow- Move down
Left Arrow - Move left
Right Arrow- Move right

Features

a. Player movement using arrow keys.
b. Coin collection system (score increases).
c. Goal-based gameplay.
d. Random maze generation for each level.
e. JSON-based data storage.
f. Object-Oriented Design implementation.

g. Modular file structure (entities, utils, storage, etc).

h. Clean UI with score display.

Screenshot

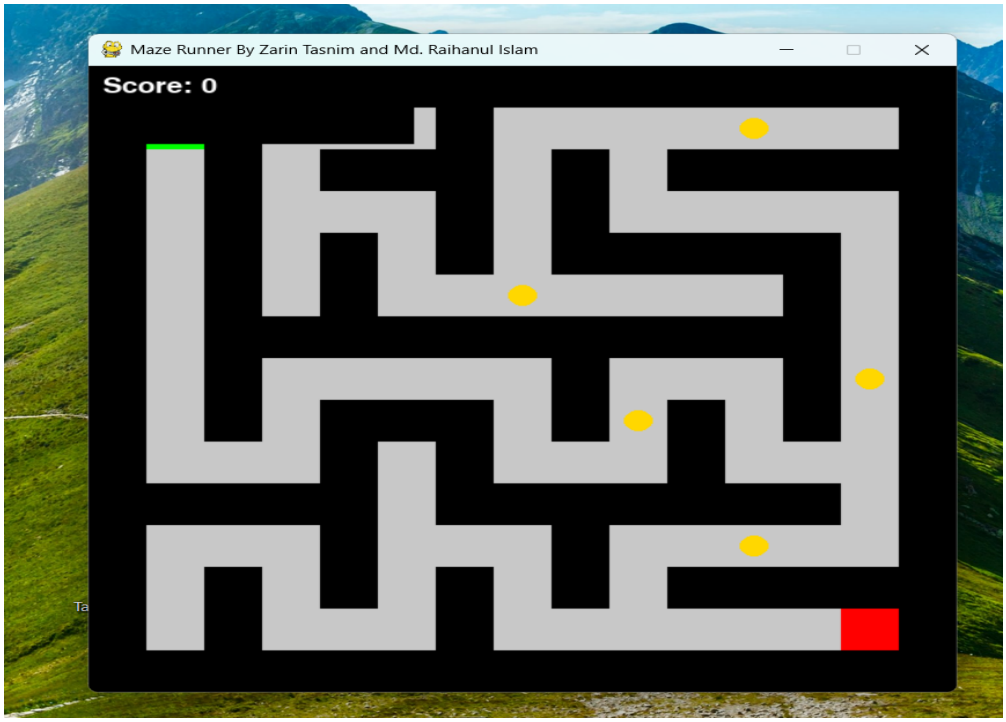

Fig 1: Maze Runner Game, start phase.

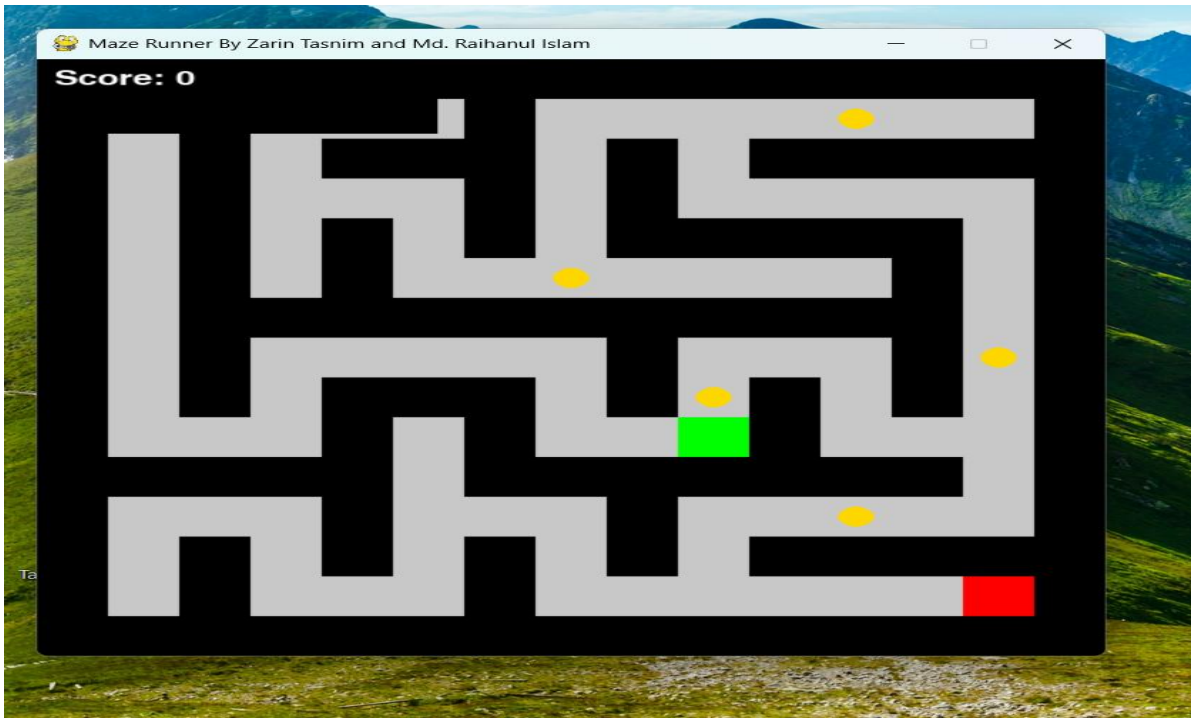

Fig 2 : Player (green box) attempt to collect coin.

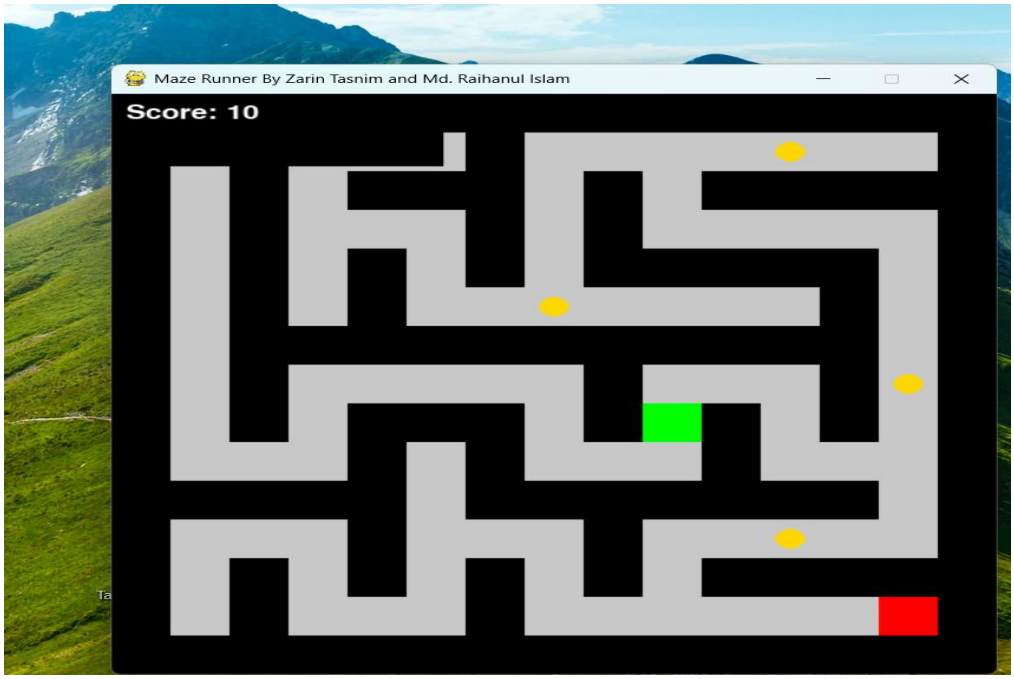

Fig 3: Player (green box) already collected the coin and score updated.

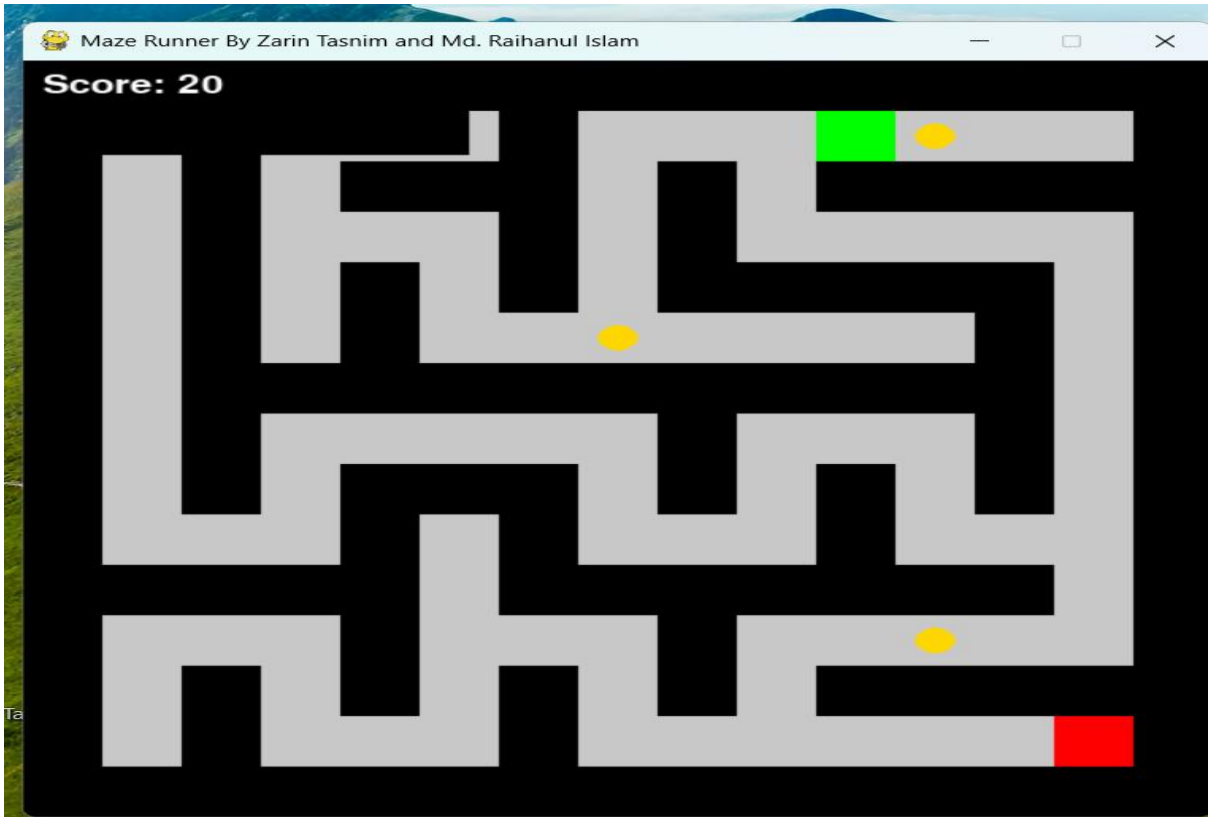

Fig 4: Player attempt to another coin.

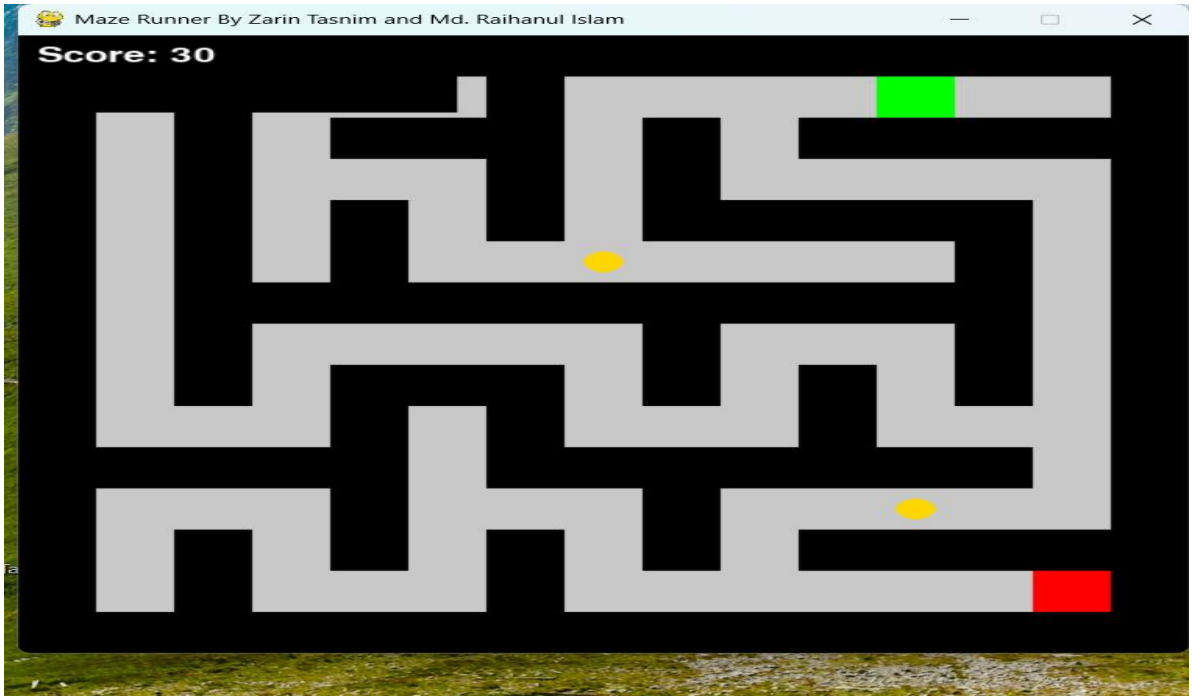

Fig 5: Player collected that coin and score is now 30.

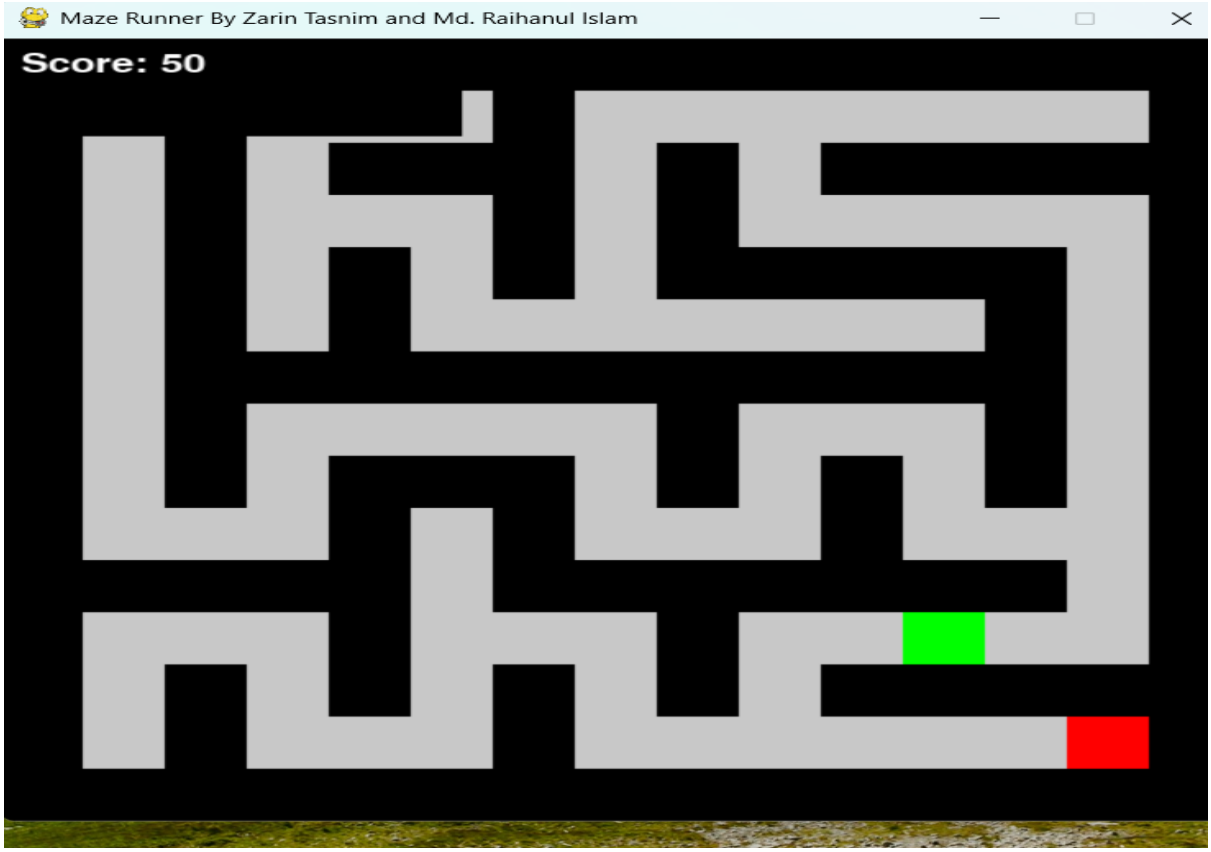

Fig 6: Player collect all coin and score is 50.

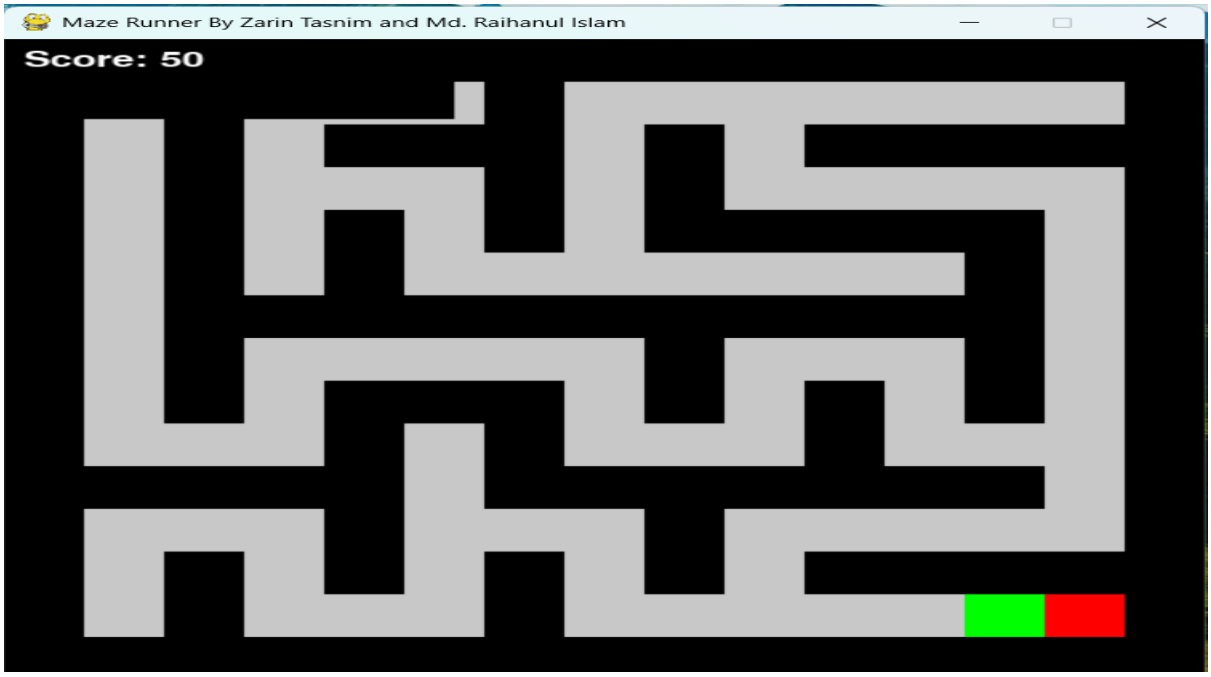

Fig 7: Game resets for next level.

Description :

The pictures show how I play the Maze Runner game. I start on the side. The Maze Runner game has coins over it. I move around using the arrow keys and pick up these coins to get points.

The goal is to reach the end, which's, on the right side of the Maze Runner game.

When I finally get to the end of the Maze Runner game I finish the level. The Maze Runner game then creates a maze.

I keep playing the Maze Runner game. It gets harder and harder as I play more of the Maze Runner game.